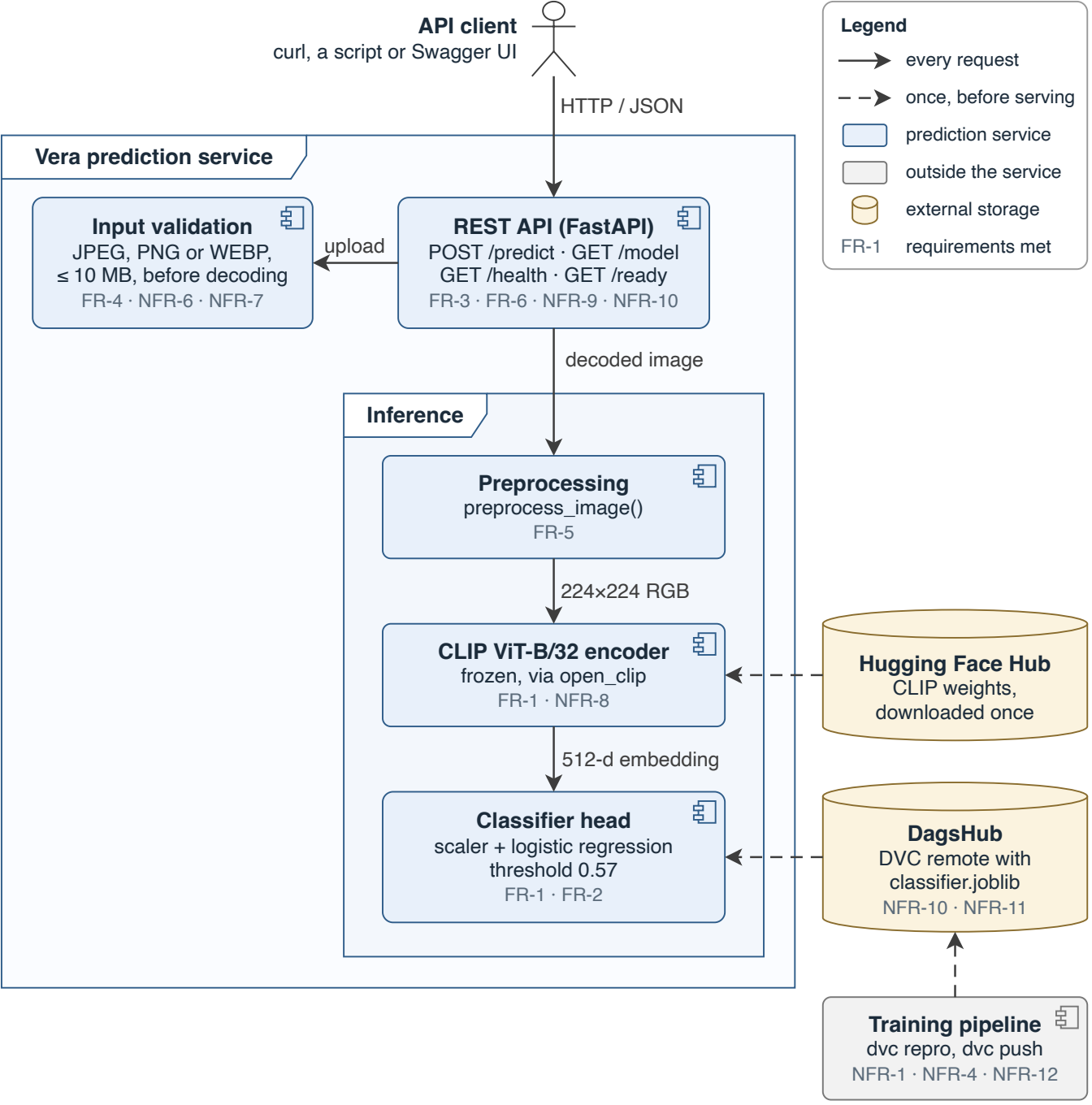
API client
curl, a script or Swagger UI
HTTP / JSON
Legend
every request
once, before serving
prediction service
outside the service
external storage
FR-1
requirements met
Vera prediction service
Input validation
JPEG, PNG or WEBP,
≤ 10 MB, before decoding
FR-4 · NFR-6 · NFR-7
upload
REST API (FastAPI)
POST /predict · GET /model
GET /health · GET /ready
FR-3 · FR-6 · NFR-9 · NFR-10
decoded image
Inference
Preprocessing
preprocess_image()
FR-5
224×224 RGB
CLIP ViT-B/32 encoder
frozen, via open_clip
FR-1 · NFR-8
512-d embedding
Classifier head
scaler + logistic regression
threshold 0.57
FR-1 · FR-2
Hugging Face Hub
CLIP weights,
downloaded once
DagsHub
DVC remote with
classifier.joblib
NFR-10 · NFR-11
Training pipeline
dvc repro, dvc push
NFR-1 · NFR-4 · NFR-12